USER INFORMATION
READ CAREFULLY

INFUSOL® M10
INFUSOL® M20
MANNITOL 10%w/v & MANNITOL 20%w/v INTRAVENOUS INFUSION BP

Product Description:
INFUSOL® M10 and **INFUSOL® M20** contain Mannitol and it is a colourless or almost colourless solution.

Composition:
Each 100mL contains:

	10%	**20%**
Mannitol	10g	20g
Water for Injection to	100mL	100mL
Osmolarity	550mOsm/L	1100mOsm/L

Pharmacodynamic:
Mannitol, hexavalent sugar alcohol which is not metabolised but eliminated via the kidneys together with a corresponding amount of water. Thus mannitol promotes urine excretion by means of osmotic diuresis also improves renal blood flow.

Pharmacokinetics:
It is a sugar (polyhydroxy aliphatic Alcohol) which, when injected intravenously, is not metabolised and is freely filtered by the glomeruli. Being non reabsorbed, it exerts considerable osmotic activity which interferes with the back diffusion of water and reabsorption of sodium in the tubules, causing osmotic diuresis.
Mannitol also acts on the Loop of Henle, where it reduces medullary blood flow. Such reduction in medullary hypertonicity leads to diminution in the reabsorption of water at the site.
As the product is administered through intravenous route it is 100% bioavailable.

Indication:
Prophylaxis of acute renal failure.
Post operative oliguria.
In forced diuresis for elimination toxic substances via the kidneys.
For decreasing intracranial pressure in case of cerebral oedema.

Recommended Dosage:
Unless otherwise prescribed
INFUSOL® M10: 500 - 1000mL per day corresponding to 50 - 100g Mannitol
INFUSOL® M20: 250 - 500mL per day corresponding to 45 - 90g Mannitol
Drop rate: 30 - 60 drops/min ≅ 90 – 180mL/hr.
The above total dose of 100g mannitol in 24 hours should only be exceeded if at least 100mL/h of urine are excreted.

or Usual adult dose :
Diuretic – Intravenous infusion, 50 to 100g as a 5 to 25% solution, administered at a rate adjusted to maintain a urine flow of at least 30 to 50mL per hour.
Cerebral oedema or Elevated intracranial pressure or Glaucoma: Intravenous infusion, 0.25 to 2grams per kg of body weight as a 15 to 25% solution, administered over a period of 30 to 60 minutes.
Note : In a small or debilitated patients, a dose of 500mg per kg of body weight may be sufficient.
Toxicity, non specific – Intravenous infusion, 50 to 200g as a 5 to 25% solution administered at a rate adjusted to maintain a urine flow of 100 to 500mL per hour
Usual adult prescribing limits: Up to 6g/kg of body weight/24hrs.

Usual pediatric dose:
Diuretic – Intravenous infusion, 0.25 to 2g per kg of body weight or 60g per square meter of body surface as a 15 to 20% solution, administered over a period of 2 to 6 hours.
Cerebral oedema or Elevated intracranial pressure or Glaucoma: Intravenous infusion, 1 to 2 g per kg of body weight or 30 to 60 g per square meter of body surface as a 15 to 20% solution, adnimistered over a period of 30 - 60 minutes.
Note : In a small or debilitated patients, a dose of 500mg per kg of body weight may be sufficient.
Toxicity, nonspecific – Intravenous infusion, up to 2g per kg of body weight or 60g per square meter of body surface as a 5 to 10% solution.

Route of administration: Intravenous (IV)

Contraindications:
Mannitol is contraindicated in patients with pulmonary congestion or pulmonary oedema.
Intracranial bleeding
Congestive heart failure
Renal failure
Dehydration

Warnings and Precautions:
The compatibility of any additives to this solution should be checked before use.
All patients given mannitol should be carefully observed for signs of fluid and electrolyte imbalance.
Single dose container.
Discard all unused content.
Do not use if leakage is detected.
If any visible solid particles, growth or turbidity appears during storage, the product should not be used.
Exposure to lower temperature may result in crystallization of mannitol. If crystallization occurs, warm up the solution to 70°C (recommeded in a hot water bath) with periodical shaking until the crystals are completely dissolved.
Cool the solution to body temperature before re-inspection and use.
Do not use unless the solution is clear.
Use filter type administration set.

Adverse Effects/Undesirable Effects:
The most serious side effect/adverse effect of mannitol is fluid and electrolyte imbalance.
Rapid administration of large dose may lead to accumulation of mannitol, over expansion of extracellular fluid, dilutional hyponatremia and occasional hyperkalemia and circulatory overload, especially in patients with acute renal failure.
Inadequate hydration or hypovolemia may be observed by the diuresis produced by mannitol, which lead to tissue dehydration, promotion of oliguria and intensification of pre-existing hemoconcentration.
Extravasation of mannitol may result in oedema and skin necrosis.

Overdose and Treatment:
Treatment of over-dosage with mannitol should be symptomatic and in particular, directed at correction of fluid and electrolyte imbalance.

Shelf Life:
5 years from manufacturing date in the proposed storage condition
Do not use after expiry.

Storage condition: Do not store above 30°C. Do not refrigerate or freeze.

Dosage form and packaging available:
INFUSOL® M10: 500mL x 20 LDPE plastic bottle per carton
INFUSOL® M20: 250mL x 20 LDPE plastic bottle per carton
500mL x 20 LDPE plastic bottle per carton

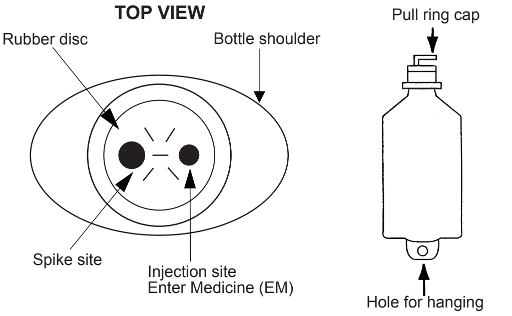
Manufacturer / Product Registration Holder:
AIN MEDICARE SDN.BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

HANDLING INSTRUCTION

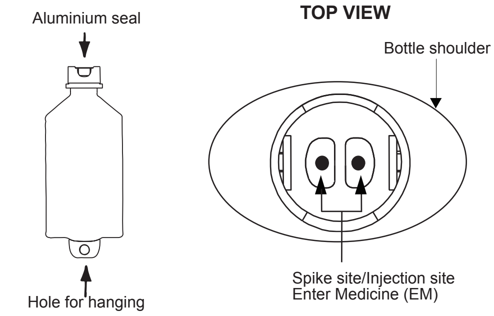
AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE

The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE



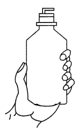
Design 1: Pull Ring Cap



Design 2: Twin Port Cap (Aluminum Seal)

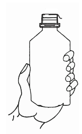
Top view after pull ring cap is removed. Please refer whichever applicable design.

INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

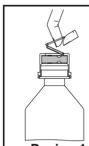


Design 1

1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.



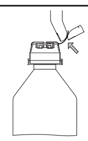
Design 2



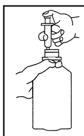
Design 1

2. Design 1: Pull the cap's ring until completely removed.

   Design 2: Pull the aluminum seal until completely removed by centre line.

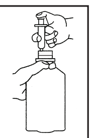


Design 2

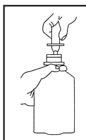


Design 1

3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.



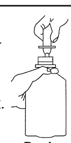
Design 2



Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 or for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set.

(Spike site is meant for spiking of the administration set and is only for single use)



Design 2

Date of Revision: 01/04/2020